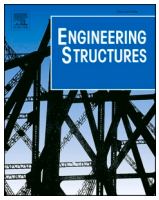
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Edge-cloud collaborative digital twin for vibration anomaly detection and localization in high-rise buildings

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ABSTRACT

Vibration accidents of high-rise buildings are frequently reported around the globe and the downtime of high-rise buildings for accident investigation can lead to substantial social and economic losses for stakeholders. This research developed an edge-cloud collaborative framework for vibration anomaly detection and vibration source localization. Firstly, a high-precision vibration sensor with wireless time synchronization and edge computing capability is introduced. Secondly, a vibration anomaly detection algorithm is proposed based on edge computing of sensor nodes. Thirdly, a vibration source localization algorithm is proposed based on cloud computing of a structural dynamics digital twin model. Finally, the effectiveness of the proposed methods is validated through a case study of a 17-story high-rise building, and the results indicate that vibration anomaly events can be reliably detected and localized. The outcomes of this study are expected to provide a useful reference for vibration anomaly management of high-rise buildings.

1. Introduction

According to the Council on Tall Buildings and Urban Habitat, there are more than 6400 high-rise buildings over 150 m built worldwide by March 1, 2023 [1]. High-rise buildings serve crucial urban functions, including hospitals and office spaces. Any sudden occurrence of vibration anomalies in these structures can impair their functionality, resulting in substantial direct and indirect economic losses [2].

Several high-rise buildings have experienced vibration anomalies that differ from those caused by typhoons or earthquakes. These vibrations are more sudden and usually cause widespread panic. It is usually impossible to find the cause of the vibration in the first place, and subsequent investigations result in prolonged closures of the buildings, leading to significant economic losses. For example, on May 18, 2021, the Saige Building (Fig. 1a) in Shenzhen encountered a vibration anomaly caused by wind-induced vortex resonance of the top-floor mast [3]. After nearly four months of closure, the building resumed operation, but the owner suffered a net profit loss of approximately 55 million RMB that year [4]. Similarly, on February 7, 2013, occupants on the 20th floor of the Information Building in Shanghai experienced intermittent abnormal shaking, prompting an emergency evacuation. Nonetheless, the cause of the building's shaking remains unknown [5]. Additionally, on September 5, 2013, a 50-story office building in Nanjing underwent several intense up-and-down shaking events within an hour due to construction work involving a road roller at a nearby construction site [6]. Moreover, on July 5, 2011, TechnoMart (Fig. 1b), a 39-story shopping mall building in Seoul, experienced abnormal vertical shaking lasting 10 min due to crowd movement inside the building, leading to a two-day evacuation of residents [7,8]. Lastly, on January 20, 2022, a 33-story high-rise residential building in Seoul encountered structural resonance caused by group dancing within the building after two months of investigation and lockdown [9].

High-rise buildings typically exhibit relatively low lateral stiffness and damping ratios, making them susceptible to external excitations such as wind loads, crowd activities, and mechanical vibrations. Structural vibrations induced by earthquakes or typhoons exhibit distinct characteristics that can be quickly identified to prevent unnecessary panic. Vibrations caused by equipment operation or human activities within the 1–10 Hz frequency range not only coincide with many high-order vibration modes of tall buildings but also fall within the sensitive

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(a) Saige Building in Shenzhen (b) TechnoMart Building in Seoul

Fig. 1. Vibration anomaly events of high-rise buildings.

range of human perception [10]. The inability to promptly determine the cause and location of such vibration sources often leads to public panic events. The aforementioned vibration anomaly events of high-rise buildings in Shenzhen, Seoul, and Nanjing demonstrate that vibration sources can be located at the top, middle, and base of the buildings. Yet, in each case, the inability to immediately identify the vibration source led to building evacuations and months-long investigations before resuming normal operations, resulting in substantial economic losses. If a vibration anomaly can be detected and localized while still weak (below the human perception threshold), timely elimination of the source can prevent such events from escalating into public incidents.

In the field of structural health monitoring (SHM) for high-rise buildings, extensive research has been conducted on operational modal analysis (OMA) [11–14] and damage detection and localization [15–17]. However, limited research has been reported on the detection and localization of vibration anomalies specifically for high-rise buildings. Existing research on vibration source localization primarily focuses on seismic localization, occupant localization, and pipeline leakage localization. For instance, travel-time inversion was proposed to determine earthquake hypocenter coordinates [18]. The double-difference earthquake location method was proposed to minimize the residuals between observed and theoretical travel-time differences of seismic pairs at each station [19]. Schloemann et al. [20] introduced a time difference of arrival (TDoA)-based method to localize impact sources within buildings. Mirshekari et al. [21] proposed an occupant localization approach using TDoA and footstep-induced floor vibrations. Lah et al. [22] introduced a pipeline leakage localization method by estimating TDoA through vibration signal correlation. Wang and Zhu [23] developed a two-stage impact source localization method based on the time of arrival (ToA) of impact waves recorded from multiple vibration monitoring points. The localization accuracy was validated through numerical simulations and field experiments conducted at a construction site with ongoing impact piling work.

These algorithms were designed for transient pulse excitations, whereas high-rise building vibration anomalies are typically non-pulse-like and persistent, limiting the applicability of TDoA/ToA methods. Consequently, vibration anomaly detection and localization in high-rise buildings remain underexplored.

In most SHM applications, the SHM system is designed based on the cloud computing pattern, where all sensor data are transmitted to a cloud server for centralized processing [24,25]. While this pattern is feasible in wired SHM systems, it has been found to be challenging and inappropriate for wireless sensor networks (WSNs) when it comes to transmitting large time-history data [26]. On one hand, implementing a WSN based on the cloud computing pattern would result in high energy consumption and reduced battery life of sensor nodes due to the continuous wireless connection to the cloud server and the significant data transmission requirements for time-history vibration data. On the other hand, the cloud computing pattern is not suitable for real-time detection of vibration anomalies due to the inherent delay in data transmission and cloud analysis.

Edge computing is a computing paradigm that performs computation at the network edge, which has gained significant research interest in the field of SHM. For instance, Wei et al. [27] proposed a multi-resource computing offload algorithm for optimizing energy consumption in mobile edge computing, resulting in more than 12 % of energy saving. Zonzini et al. [28] introduced an innovative extreme edge data compression strategy tailored for low-power and low-cost SHM applications. Experimental results from wind turbine monitoring demonstrated that this strategy reduced the energy consumption of a sensor node by over 16 %. Additionally, Abner et al. [29] presented three strategies for conserving battery power in wireless monitoring sensors for infrastructures using edge computing, significantly enhancing the sensor's battery lifespan and reducing the frequency of on-site hardware maintenance. Based on the above research, edge computing shows promise in addressing challenges related to response time, battery life, and bandwidth costs [30]. However, due to limited computing and memory resources, as well as network bandwidth constraints, edge computing faces challenges in achieving satisfactory performance when dealing with complex problems.

Owing to the complementary advantages and disadvantages of cloud computing and edge computing, an increasingly common approach is to integrate them in real-world applications [31]. This approach is known as edge-cloud collaboration (ECC), which combines the low latency characteristics of edge computing with the powerful computing capabilities of cloud computing [32]. Early explorations of ECC have been conducted in both academia and industry. For example, Yu et al. [33] designed an ECC framework for industrial anomaly detection, where cloud algorithms are responsible for extracting and optimizing anomaly diagnosis rules, while edge algorithms detect anomalies based on these rules. This method significantly reduces network load and computational pressure on the cloud. Gang et al. [34] proposed an ECC framework for electric load monitoring, which performs processing at the edge and optimization in the cloud. Ding et al. [35] introduced an ECC framework for healthcare monitoring that enables lesion detection at the edge and high-performance modeling in the cloud. As observed, current ECC frameworks primarily operate in the mode of edge detection with cloud-based dataset training or modeling. However, no reports have been made on edge-cloud collaborative intelligence for anomaly detection and localization in high-rise buildings.

The aforementioned edge and clouding computing can help to acquire, process and analysis data from physical world. When combined with more complicated simulation and visualization models as the digital replica of the real world entities, the concept of digital twin can be involved. A digital twin represents a cyber-physical system that enables visualization, modeling, simulation, analysis, prediction, and optimization [36]. It has found applications in various civil engineering fields, including buildings [37–40], infrastructure [41,42], and construction sites [43,44]. The digital twin is considered a promising technology with the potential to transform the construction industry and provide solutions to various challenges [45]. Furthermore, Jiang et al. [46] argued that technologies such as digital twin, building information modeling (BIM), and cyber-physical systems (CPS) are not mutually exclusive; instead, they can mutually reinforce civil engineering digitalization. Therefore, the digital twin represents a broader concept that can be integrated with edge-cloud collaborative computing, BIM models, and simulation models to establish a comprehensive digital system for enhanced management and disaster prevention of civil structures.

Therefore, an edge-cloud collaborative digital twin framework is presented for the purpose of vibration anomaly detection and localization in high-rise buildings. The composition of the framework is detailed in Section 2, followed by the introduction of the hardware composition and vibration anomaly detection algorithm in Section 3. In Section 4, a vibration source localization method is proposed. Additionally, Section

5 introduces a BIM-based online monitoring and management platform. To demonstrate the effectiveness of the framework, a case study is conducted for a 17-story teaching building in Section 6.

2. Overview of the methodology

The proposed edge-cloud collaborative (ECC) digital twin framework for vibration anomaly detection and localization in high-rise buildings comprises three main components: edge sensing for vibration anomaly detection, cloud computing for vibration source localization, and a BIM-empowered monitoring management platform, as illustrated in Fig. 2.

The first module is edge sensing for vibration anomaly detection, which involves: (1) a novel wireless time-synchronized smart sensor node, and (2) a vibration anomaly detection algorithm specifically designed for high-rise buildings.

The second module utilizes cloud computing for vibration source localization and BIM-based visualization, and consists of three components: (1) a digital twin model for structural dynamic simulation, constructed using modal parameters derived from operational modal analysis; (2) a vibration anomaly detection algorithm that performs vibration source localization by utilizing both the measured and simulated spectrum amplitude features; and (3) a BIM-empowered online monitoring and management platform capable of displaying real-time vibration data during normal operation states and visualizing abnormal vibration sources through the BIM model during abnormal states.

3. Edge sensing for vibration anomaly detection

The detection of vibration anomaly in high-rise buildings requires high-precision time-synchronized monitoring of building vibrations. In this section, details of the smart sensor node and vibration anomaly detection algorithm are introduced.

3.1. Wireless time-synchronized smart sensor

Extensive research has been conducted on wireless time-synchronized smart sensors for structural health monitoring, with notable contributions including the state-of-the-art MEMS-based Xnode sensor developed by Professor Spencer's team [47]. This sensor incorporates three dedicated circuit boards handling data acquisition, wireless transmission, and power management respectively. The structural health monitoring system consists of multiple sensing nodes and a gateway node, with wireless time synchronization implemented using the IEEE 802.15.4 protocol [48]. This system has been successfully deployed in bridge monitoring applications.

Building upon the Xnode design concept, we propose a new wireless time-synchronized smart sensor (WTSS) featuring three key improvements: First, our design achieves greater functional integration by

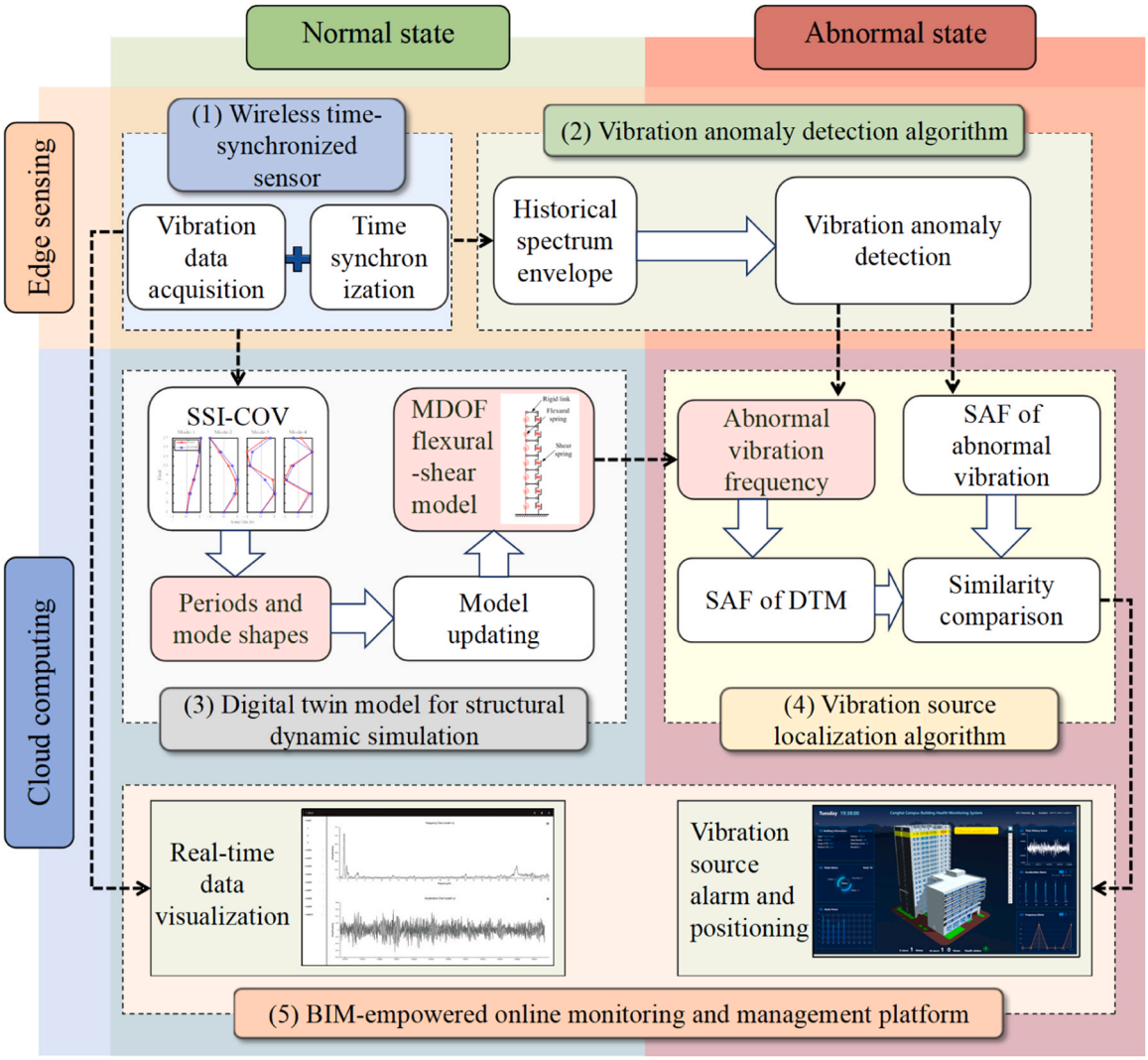


Fig. 2. Framework of proposed methodology.

combining multiple capabilities including data acquisition, wireless transmission, power management and edge computing into a compact circuit layout (Fig. 3a); second, the nodes implement LoRa-based wireless time synchronization, which enables communication over long distance between nodes; third, our system eliminates the need for gateway nodes. Each node directly transmits data to the cloud via 4 G wireless communication, which not only removes the requirement for separate gateway nodes but also provides substantially greater bandwidth for cloud connectivity.

The logical composition of the WTSS is illustrated in Fig. 3b. The node primarily comprises a motherboard, an acceleration sensor module, a Raspberry Pi computing module, a LoRa wireless communication module, and a 4 G wireless transmission module. The motherboard integrates an STM32 microcontroller chip (ARM Cortex-M), a LoRa SX1276 module, and an acceleration sensor. The STM32 performs two primary functions: First, it acquires vibration data from the sensor module through the serial peripheral interface (SPI) and transmits these data to the Raspberry Pi via a serial port. Second, it controls the LoRa wireless communication module for time synchronization. The LoRa SX1276 module, manufactured by Semtech [49], is an ultra-long-range, highly sensitive wireless communication module. For wireless time synchronization among WTSS nodes, we implement a chain network structure comprising two node types: a single root node R and multiple relay nodes J_i. The root node R is positioned at the building's top floor, with time synchronization timestamps propagating between adjacent nodes in a top-down manner along the chain network (Fig. 3c). This synchronization between root node R and relay node J_i, as well as between relay nodes, is implemented through timestamp exchange using the Network Time Protocol (NTP) [50].

For vibration data acquisition, the study utilizes the EPSON M-A352AD10 acceleration sensor, a tri-axial digital accelerometer known for its high sensitivity and low noise level. The sensor has a typical noise density of $0.2ug/(Hz)^{1/2}$ [51].

The Raspberry Pi Zero 2 W module [52], equipped with a quad-core 64-bit ARM Cortex-A53 CPU, is utilized for edge computing. The Raspberry Pi board runs a Linux operating system (64-bit, Debian 11.2 version), which incorporates a Python language compiler. By deploying vibration signal processing algorithms written in Python, the system achieves the identification of abnormal vibrations, with the detailed implementation process of the anomaly detection algorithm elaborated in Section 3.2. The vibration data and analysis results from the WTSS node are transmitted to the cloud server via the Waveshare SIM7600G 4 G module [53]. The data transmission employs the MQTT protocol [54], where the sensor node acts as the publisher under QoS1 level, sending the collected data through an established connection to a designated topic on the cloud platform. The cloud platform, functioning as the subscriber, continuously monitors the topic to receive real-time data reported by the sensor, thereby completing the data transfer from the terminal device to the cloud.

3.2. Vibration anomaly detection algorithm

This section describes the proposed vibration anomaly detection algorithm implemented at the sensor node, as depicted in Fig. 4a. The algorithm comprises four primary steps:

(1) The WTSS nodes are adopted to acquire the vibration on different stories of a high-rise building as the historical data for a certain period of time. Note that the historical vibration data is acquired during normal operation of the building, the acquired historical vibration data are deemed as the baseline to evaluate the normality of current vibration.

(2) Short-time Fourier transform (STFT) is performed on the historical vibration data using the window size of a few minutes. The maximum amplitude values at different frequency points are extracted as the historical spectrum envelope, represented by the dashed line in Fig. 4b.

(3) Applying STFT to the current vibration data using the same window size to obtain the current spectrum, illustrated by the red line in Fig. 4b.

(4) Comparing the current spectrum with the historical spectrum envelope. If the current spectrum exceeds α times the historical spectrum envelope at any frequency point, it is determined that a vibration anomaly event has occurred, and the corresponding frequency is recorded as the vibration anomaly frequency ω.

To ensure a timely response for vibration anomaly detection, the algorithm can be implemented using Python and executed on the Raspberry Pi module of the WTSS. When a vibration anomaly event is detected, the WTSS nodes promptly upload the spectrum feature data, which includes the vibration anomaly frequency ω and its amplitude, along with the high-sampling-rate vibration time-history data to the cloud server.

4. Cloud computing for vibration source localization

4.1. Digital twin model for structural dynamic simulation

In this section, the structural dynamics digital twin model (DTM) of high-rise buildings is established to assist in finding the source of excitation of a vibration anomaly event. The DTM is constructed through model updating based on the measured frequency spectrum data. The process of developing the DTM involves model establishment, operational modal analysis, and model updating. This paper specifically focuses on reinforced concrete (RC) frame-shear wall high-rise buildings, which can be effectively modeled using a multiple degrees of freedom (MDOF) flexural-shear model [55]. The MDOF flexural-shear model is commonly employed for the design and simulation of RC frame-shear wall buildings, with shear wall elements represented by Euler-Bernoulli springs and frame elements modeled as shear springs. Previous studies have demonstrated the performance of the MDOF flexural-shear model in simulating the dynamic responses of RC

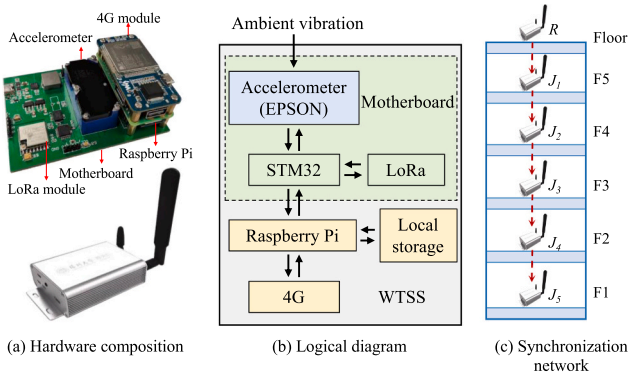
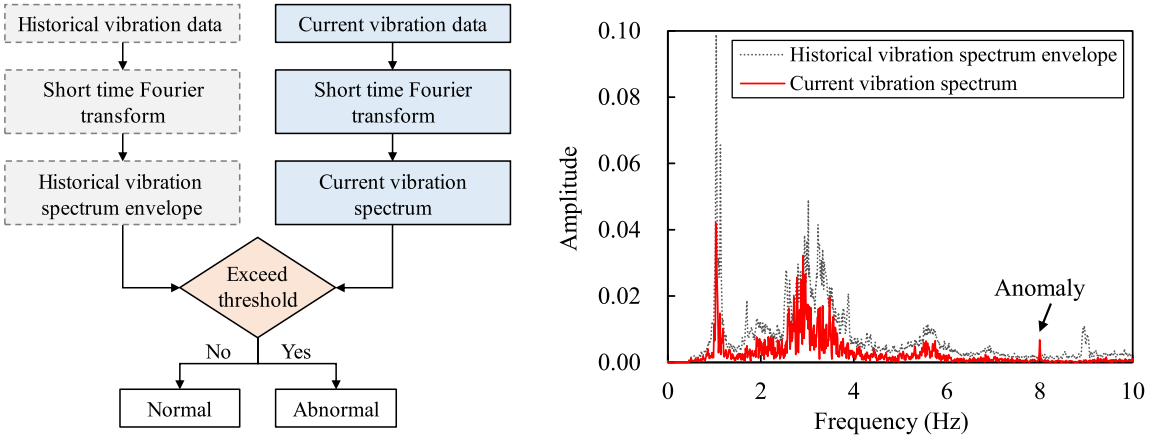


Fig. 3. Hardware composition of the WTSS node.



(a) Procedure of vibration anomaly detection

(b) Historical spectrum envelope and current spectrum

Fig. 4. Proposed algorithm for vibration anomaly detection.

frame-shear wall buildings [56–59]. The global stiffness matrix K and mass matrix M of a representative three story MDOF flexural-shear model is demonstrated in Eqs. (1) and (2).

$$K=\begin{bmatrix} \frac{12i_1}{l_1^2}+\frac{12i_2}{l_2^2}+\Psi_1+\Psi_2 & -\frac{12i_2}{l_2^2}-\Psi_2 & 0 & \frac{6i_2}{l_2} & -\frac{6i_1}{l_1}\frac{6i_2}{l_2} & 0 \\ -\frac{12i_2}{l_2^2}-\Psi_2 & \frac{12i_2}{l_2^2}+\frac{12i_3}{l_3^2}+\Psi_2+\Psi_3 & -\frac{12i_3}{l_3^2}-\Psi_3 & -\frac{6i_2}{l_2}\frac{6i_3}{l_3} & -\frac{6i_2}{l_2}\frac{6i_3}{l_3} & \\ 0 & -\frac{12i_3}{l_3^2}-\Psi_3 & \frac{12i_3}{l_3^2}+\Psi_3 & 0 & -\frac{6i_3}{l_3} & -\frac{6i_3}{l_3} \\ \frac{6i_2}{l_2} & -\frac{6i_1}{l_1} & -\frac{6i_2}{l_2} & 0 & 4(i_1+i_2) & 2i_2 \quad 0 \\ \frac{6i_2}{l_2}\frac{6i_3}{l_3} & -\frac{6i_2}{l_2} & -\frac{6i_3}{l_3} & 2i_2 & 4(i_2+i_3) & 2i_3 \\ 0 & \frac{6i_3}{l_3} & -\frac{6i_3}{l_3} & 0 & 2i_3 & 4i_3 \end{bmatrix} \quad (1)$$

$$M=\begin{bmatrix} m_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & m_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & m_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_3 \end{bmatrix} \quad (2)$$

where $i_j=\Pi_j/l_j$, Π_j is the total flexural stiffness of the jth floor, l_j is height of the jth floor; Ψ_j is the total shear stiffness of the jth floor; m_j is the mass of the jth floor, and I_j is the moment of inertia of flexural members on the jth floor.

The l_j, m_j, and I_j in Eqs. (1)–(2) can be determined based on basic design information of the building. The initial flexural and shear stiffness parameters Π_j and Ψ_j can be determined according to the first and second mode frequencies [55]. Subsequently, the flexural and shear stiffness parameters can be further updated according to the modal parameters obtained through operational modal analysis.

Specifically, the procedure for establishing the DTM consists of four

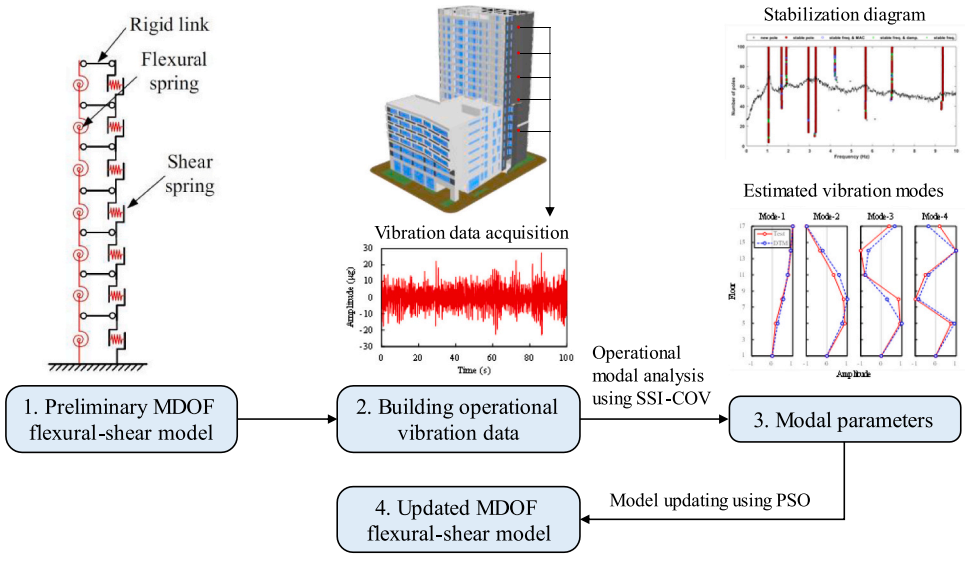


Fig. 5. Digital twin model for structural dynamic simulation using MDOF flexural-shear model.

steps (Fig. 5): (1) Creating a preliminary MDOF flexural-shear model based on the basic design information [55,56], which includes structural height, floor mass, first and higher-order natural frequencies, etc. (2) Collecting vibration data from the building during normal operation using WTSS nodes. (3) Applying the covariance-driven stochastic subspace identification method (SSI-COV) [60] to perform operational modal analysis and extract the modal parameters such as building natural frequencies and mode shapes. (4) Using the obtained modal parameters as targets, the stiffness parameters Π_j and Ψ_j of the MDOF flexural-shear model can be updated using the particle swarm optimization algorithm (PSO) [61], and the loss function is presented in Eq. (3).

$$L = \alpha \sum_{k=1}^{p} \frac{1}{p}[(\omega_k - \widehat{\omega}_k)/\omega_k]^2 + \sum_{k=1}^{p}\sum_{j=1}^{q} \frac{1}{pq}[(\varphi_{jk} - \widehat{\varphi}_{jk})/\varphi_{jk}]^2 \quad (3)$$

where ω_k and $\widehat{\omega}_k$ are the actual and predicted natural frequencies of the kth mode; p is the number of modes adopted in the model updating; ϕ_{jk} and $\widehat{\varphi}_{jk}$ are the actual and predicted mode shape parameters of the kth mode on the jth floor; q is the number of floors; α is a factor that can influence the contribution of the first and second terms.

4.2. Vibration source localization algorithm

The vibration anomaly investigated in this study primarily focuses on building vibrations caused by harmonic excitation. A series of harmonic excitation tests conducted in this study reveal that the amplitude of structural frequency response at different floors varies significantly when horizontal harmonic excitation is applied at a certain floor. Fig. 6 illustrates the acceleration spectra of the 5th, 8th, 11th, 14th, and 17th floors of a 17-story high-rise building when horizontal 8 Hz harmonic excitation was applied to the 5th, 8th, and 11th floors, respectively. The results indicate significant variations in the amplitudes of the acceleration spectra at 8 Hz for the five floors.

As evident in Fig. 6, the floor closest to the excitation source does not necessarily exhibit the largest amplitude. This is because in high-rise buildings, the dynamic response typically results from the superposition of multiple vibration modes. Excitation at different floors activates distinct modal combinations (Fig. 7). When the excitation source is proximate to the stationary point of a vibration mode, the floor near the stationary point exhibits relatively small amplitude.

The DTM discussed in Section 4.1 is utilized for conducting time-history analysis. An 8 Hz harmonic excitation is applied to the DTM, and the acceleration spectra of all stories are obtained. The amplitudes at 8 Hz in the acceleration spectra from both the test and the simulation are extracted as spectrum amplitude feature (SAF) vectors. Fig. 8 illustrates the normalized SAF vectors for different excitation conditions. The comparison exhibits correlative trends between the SAF vectors obtained from the test and the simulation under identical harmonic excitation conditions. Additionally, the SAF vectors exhibit distinct patterns for excitations on different floors, allowing for the prediction of the excitation location based on the shape of the obtained SAF vector.

To further demonstrate this phenomenon, we have added a simulation case study where 8 Hz harmonic excitations were sequentially applied on the 2nd to 17th floors. As shown in Fig. 9, the SAFs of the building exhibits completely different patterns when excited at different floors. By comparing the measured SAF with the simulated SAF values, the approximate location of the vibration source can be determined. Moreover, the results in Fig. 9a (pink curve) further demonstrate the test observations in Fig. 6. When the excitation was applied on the 5th floor, the large vibrational amplitudes occurred on the 8th and 14th floors, while the 5th and 11th floors showed relatively small responses. Consequently, an algorithm for vibration source localization is proposed based these observations.

The fundamental idea of the proposed vibration source localization method is to compare the SAF vector of the measured abnormal vibration with the SAF vectors obtained through the structural dynamic simulation of the DTM under all possible excitation load cases. The method consists of six steps, as depicted in Fig. 10:

(1) Collect the frequency spectra of all WTSS nodes after receiving the vibration anomaly alert from the edge nodes.

(2) Compute the normalized (L^2-Norm = 1) SAF vector v (Eq. (4)) of the measured vibration anomaly based on the amplitude of the abnormal frequency ω.

(3) Perform time-history analyses of the DTM with unit sinusoidal

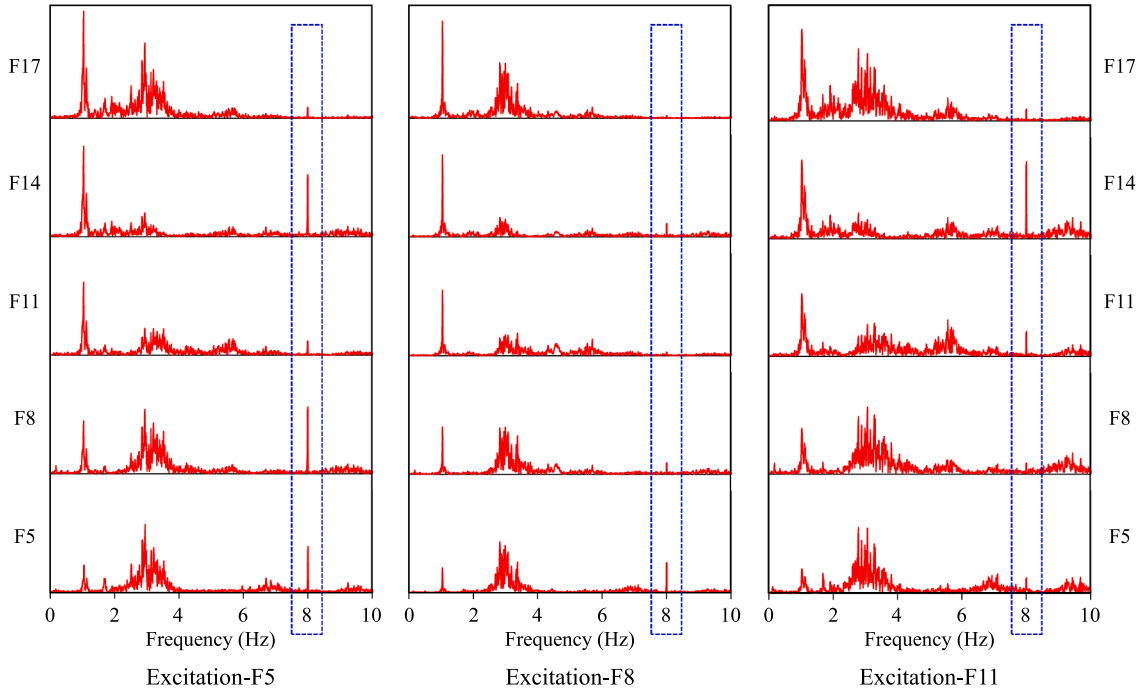


Fig. 6. Acceleration spectra of different floors under 8 Hz excitation on the 5th, 8th and 11th floors respectively.

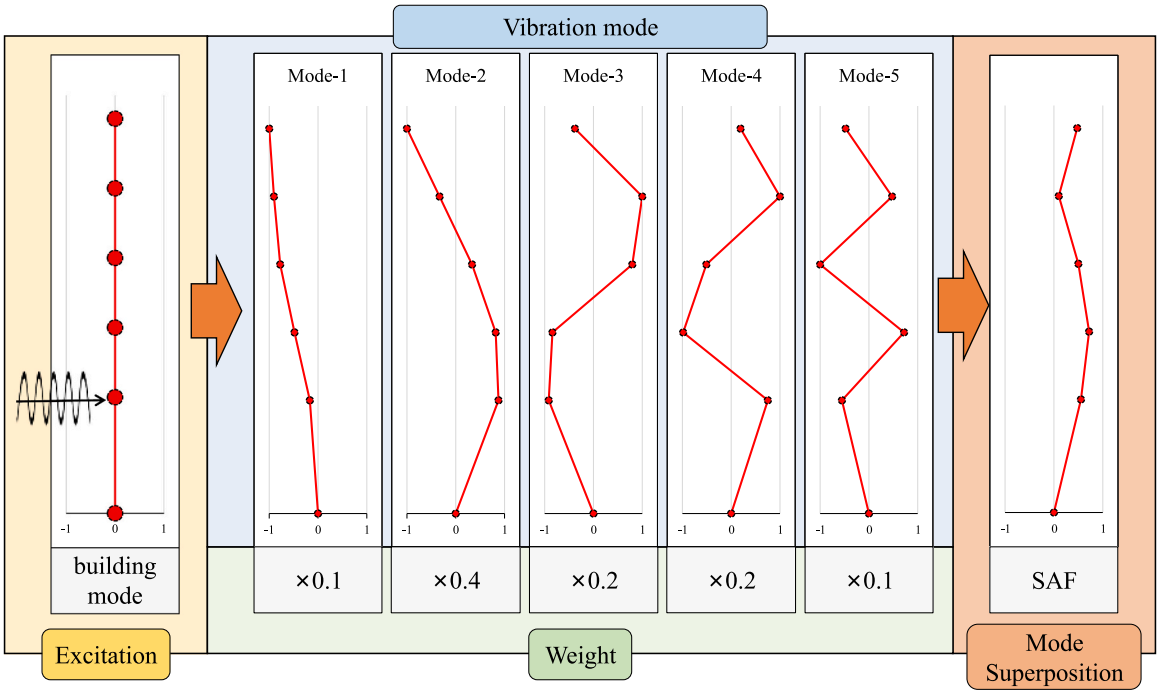


Fig. 7. Schematic diagram of dynamic response superposition of multiple vibration modes.

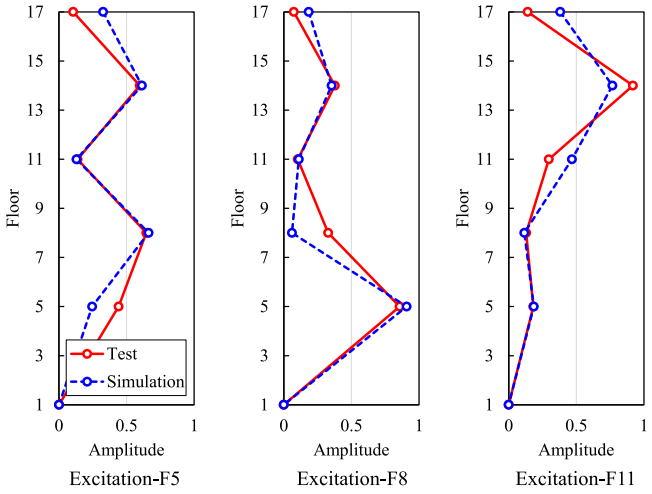


Fig. 8. Test and simulated SAF vectors with excitation on the 5th, 8th and 11th floors respectively.

excitation applied to different floors, respectively.

(4) Calculate the normalized SAF matrix W (L^2-Norm = 1 for each column vector) (Eq. (5)) based on the amplitude of the frequency spectra of the DTM subjected to different load cases.

(5) Determine the similarity between the measured SAF vector v and the simulated SAF matrix W using Eq. (6).

(6) Calculate the floor of the vibration source, l, by identifying the argument of the maxima in the similarity vector s according to Eq. (7).

$$v = [\rho_1 \rho_2 \ldots \rho_j] \tag{4}$$

$$W = [\varphi_1 \varphi_2 \ldots \varphi_k] = \begin{bmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1k} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ \varphi_{j1} & \varphi_{j2} & \cdots & \varphi_{jk} \end{bmatrix} \tag{5}$$

$$s = v \times W = [\rho_1 \rho_2 \ldots \rho_j] \times \begin{bmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1k} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ \varphi_{j1} & \varphi_{j2} & \cdots & \varphi_{jk} \end{bmatrix} = [s_1 s_2 \ldots s_k] \tag{6}$$

$$l = \operatorname{argmax}(s) \tag{7}$$

where ρ_j is normalized spectral value of the measured anomaly vibration frequency on the jth floor; φ_{jk} is the normalized spectral value of the simulated response vibration on the jth floor of the DTM with excitation subjected on the kth floor; s_k is the similarity value between the measured SAF vector v and the simulated SAF vector φ_k with excitation subjected on the kth floor.

5. BIM-empowered online monitoring and management platform

This study has developed a web-based platform for data visualization and vibration anomaly management in high-rise buildings. The platform specifically integrates the BIM model of the building to visualize the sensor locations and the floors where vibration sources are detected. Fig. 11 illustrates the main components of the platform, which includes the data flow module, the back-end data processing module, and the front-end visualization module.

(1) Data flow module:

The data flow module of the platform operates as follows. The frequency spectra and vibration data from all WTSS nodes are transmitted to the cloud server via the MQTT protocol. The server stores the newly uploaded data in the MySQL database as hot data. After a certain time period, the server transfers the hot data from the MySQL database to object storage as cold data to improve operational efficiency. The cold data includes raw data, which preserves the original vibration data with a high sampling rate, and downsampled snapshot data for fast preview.

(2) Back-end data processing module:

The back-end data processing module of the platform connects with the MySQL database and object storage using the Flask-SQLAlchemy library [62]. The module consists of a series of algorithms, including operational modal analysis and DTM updating using the obtained modal

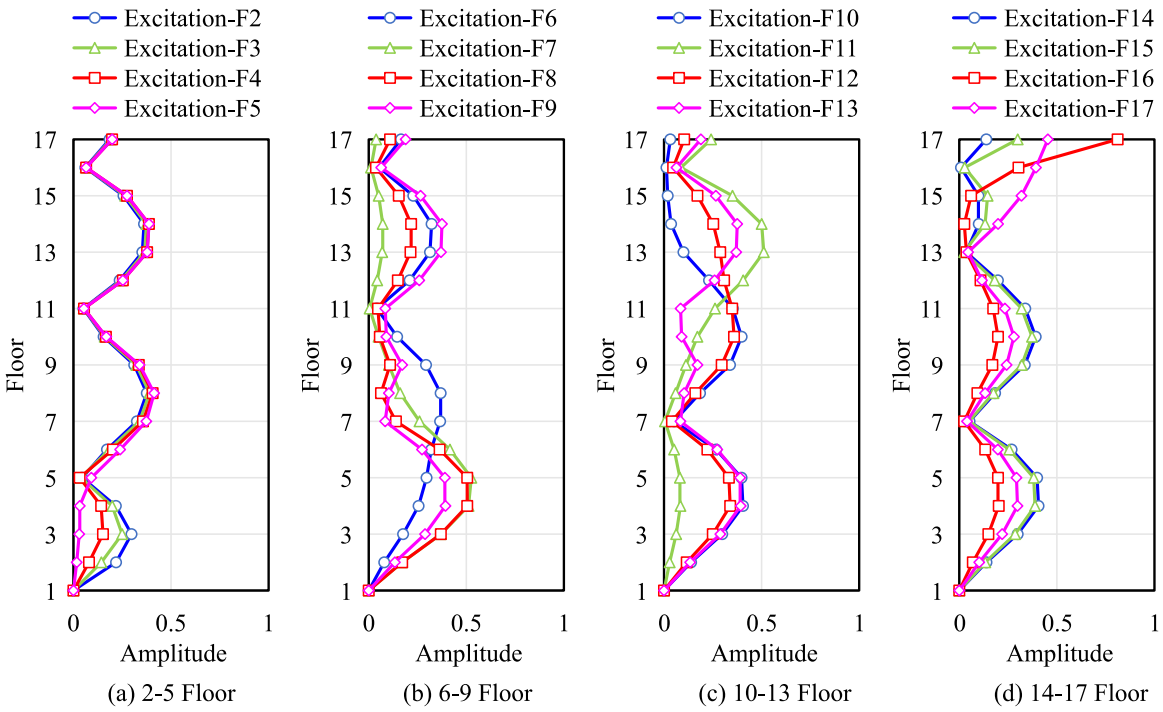


Fig. 9. Simulated SAF vectors with excitation on the 2nd to 17th floors respectively.

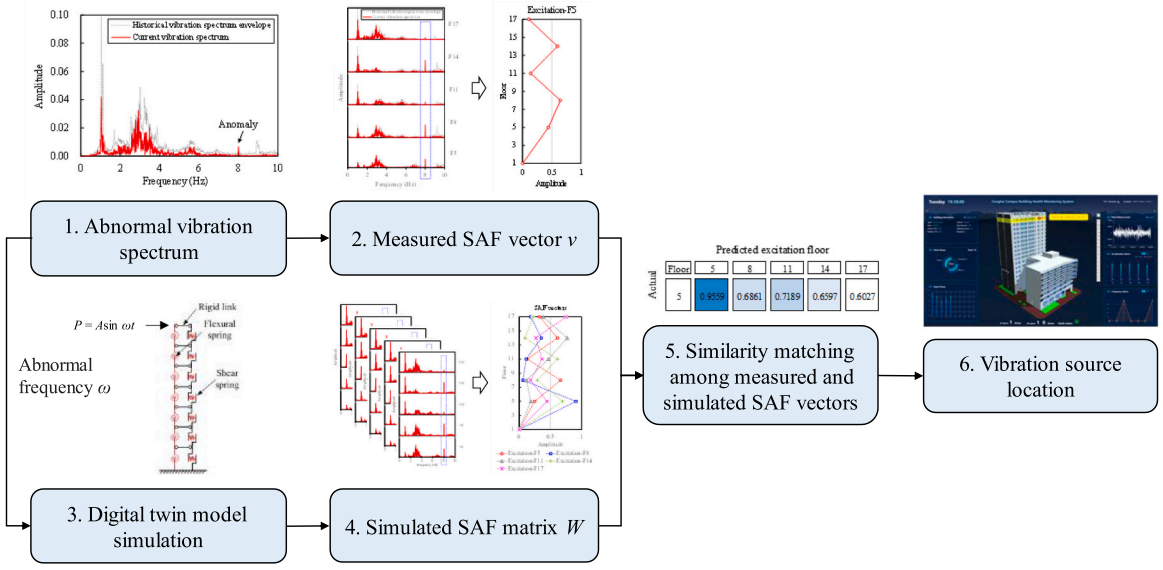


Fig. 10. Procedure of the vibration source localization method.

parameters. The module transmits the analysis results to the front-end visualization module through the WebSocket protocol for real-time display.

(3) Front-end visualization module:

The front-end visualization module of the platform operates as follows. The BIM model of the building is pre-processed to reduce unnecessary components for fast buffer and maneuver on the web-based platform. The BIM model is converted to the glb format and imported to the platform using the Three.js engine [63]. The module decodes the data received from the back-end through the WebSocket protocol, enabling real-time display of acceleration time-history data, frequency spectrum data, and vibration anomaly alert events.

6. Illustrative examples

To demonstrate the effectiveness of the proposed ECC digital twin framework, a 77.7-meter-high, 17-story RC frame-shear wall teaching building is studied in this section. Ambient vibration tests were performed to assess the performance of the WTSS node. Additionally, a series of sinusoidal excitation tests were conducted to illustrate the effectiveness of the vibration anomaly detection and localization methods.

6.1. Performance of the WTSS

In this study, the vibration measurement performance of the proposed WTSS was verified by comparing it with a high-precision wired

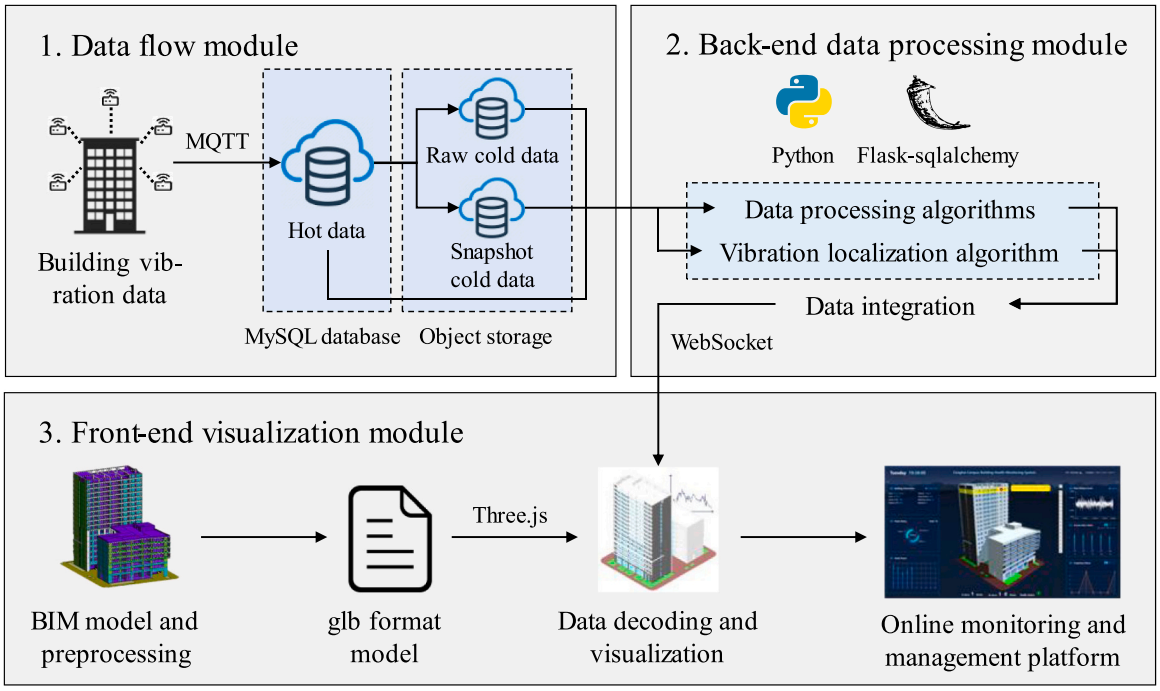


Fig. 11. Schematic diagram of the online monitoring platform.

sensor. As depicted in Fig. 12, the WTSS was compared to the wired piezoelectric sensor PCB393b31 (PCB31), which was installed on the 17th floor of the building. PCB31 is a single-axis accelerometer that was horizontally mounted on a test stand, aligned with the weak axis direction of the building.

The sampling rate of the WTSS was set to 500 Hz, while the PCB31 sensor utilized the NI9234 acquisition system with a sampling rate of 1651.6 Hz, which is the lowest setting available for the system. Both sensors collected ambient vibration data for a duration of 3 min. The collected data underwent processing, including detrending, band-pass filtering (0.2–20 Hz), and fast Fourier transform (FFT), to derive the acceleration time-history data and Fourier spectra. As illustrated in Fig. 13a-b, in the time domain, the acceleration waveforms measured by the two sensors closely resemble each other, exhibiting an amplitude range of ±35 μg. In the frequency domain (Fig. 13c-d), the spectra of the two sensors also exhibit good agreement, with the first peak corresponding to the frequency of 1.05 Hz, which represents the first-order natural frequency in the direction of the weak axis of the building. This test demonstrates that the WTSS can achieve precise and reliable vibration measurement performance comparable to that of the high-precision PCB sensor.

To validate the time synchronization accuracy of the monitoring network, multiple WTSS nodes were deployed across multiple floors in the studied high-rise building. As illustrated in Fig. 14a, the WTSS nodes implemented time synchronization within the chain network using the procedure presented in Section 3.1. Meanwhile, each WTSS node was externally connected to a global navigation satellite system (GNSS) module for receiving pulse-per-second (PPS) signals from satellites. By comparing local timestamps recorded upon reception of each PPS signal, the clock offset between WTSS nodes was recorded.

As illustrated in Fig. 14b, each WTSS was mounted with a GNSS antenna to ensure uninterrupted satellite signal reception. The performance of the proposed LoRa-based time synchronization method was evaluated with 300-s and 600-s intervals, and each test was conducted continuously for 6000 s. The results presented in Fig. 15 demonstrate consistent synchronization errors of less than 20 μs across both test scenarios, with maximum observed deviations of 16 μs (300-s interval) and 19 μs (600-s interval). These findings quantitatively confirm the reliability of the proposed wireless time synchronization method for building-scale deployment.

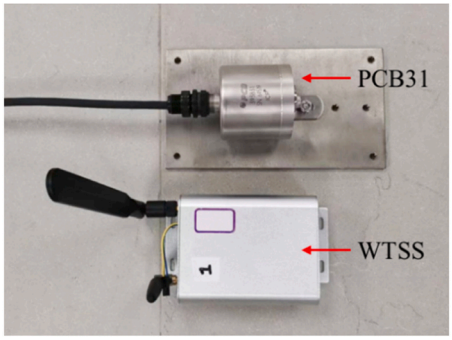


Fig. 12. Ambient vibration test on the 17th floor of the teaching building.

6.2. *Vibration anomaly detection and localization*

The tests conducted in the previous section have successfully verified the performance of the proposed WTSS. This section focuses on demonstrating the effectiveness of the vibration anomaly detection and localization methods. Because previous abnormal vibration events were predominantly caused by steady-state single-frequency vibrations, including low-frequency mechanical reciprocations and mast oscillations, harmonic excitation was adopted as the abnormal excitation. Specifically, load case of vibration anomaly in the building is simulated through the sinusoidal motion of a mass block driven by a shaking table, as shown in Figure 16a. The mass block is 21 kg, the measured acceleration time history of the mass is shown in Figure 16b. The peak acceleration is 0.403 g and the calculated excitation force amplitude is 82.9 N. The excitation force is transferred to the building structure through the friction between the shaking table and the slab.

The arrangement of the shaking table and WTSS nodes in the building is shown in Figure 17. Due to the prohibitive costs of full-floor instrumentation in real-world applications, especially for high-rise and super-tall structures, a sparse sensor deployment strategy is adopted,

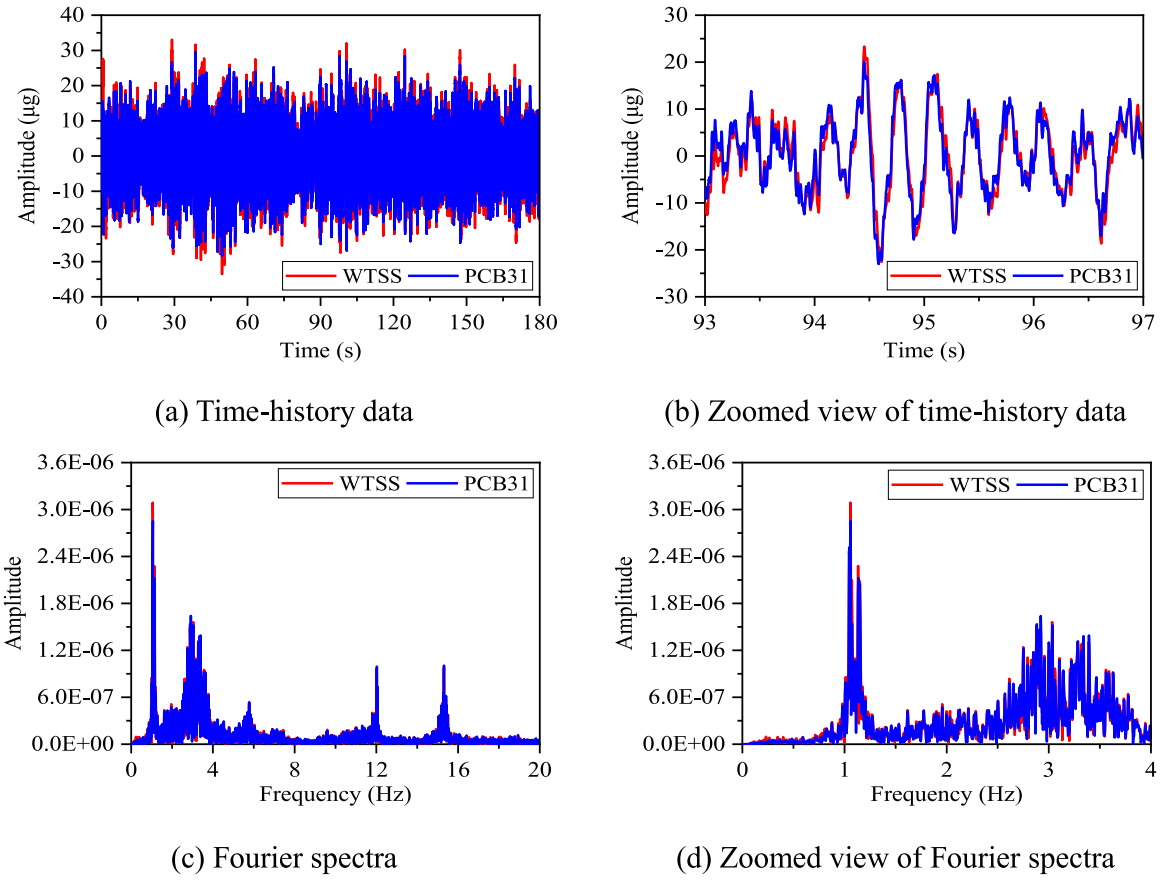


Fig. 13. Comparison of time-history and spectrum data.

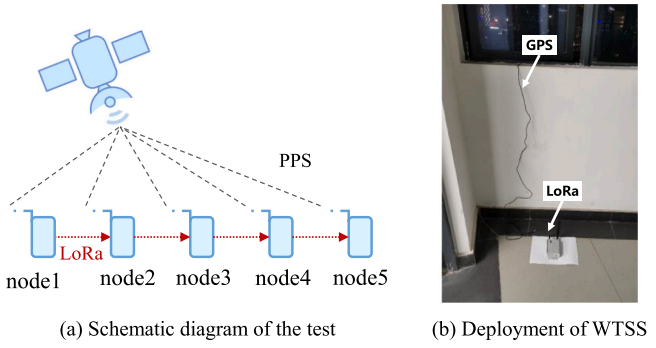


Fig. 14. Synchronization performance test using GNSS PPS as reference.

with WTSS-1 to WTSS-5 arranged on the 17th, 14th, 11th, 8th and 5th floors respectively, constituting a five-node vertical chain sensor network. The shaking table is arranged along the x-axis of the corridor on the 5th, 8th, 11th, 14th and 17th floors, respectively, to simulate the vibration anomaly on a specific floor.

6.2.1. *Digital twin model updating*

During routine operation, acceleration time-history records were continuously streamed to the cloud server, where the covariance-driven stochastic subspace identification (SSI-COV) algorithm [60] was applied to extract the structure's modal properties. The resulting stabilization diagram (Fig. 18) serves to eliminate spurious modes, where the red poles satisfy three stability criteria: (1) frequency variation between adjacent poles less than 1 %, (2) damping ratio variation less than 5 %, and (3) modal assurance criterion (MAC) values greater than 0.98 for all

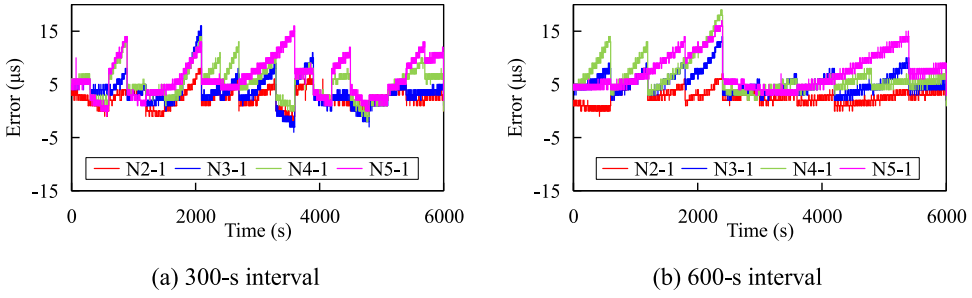
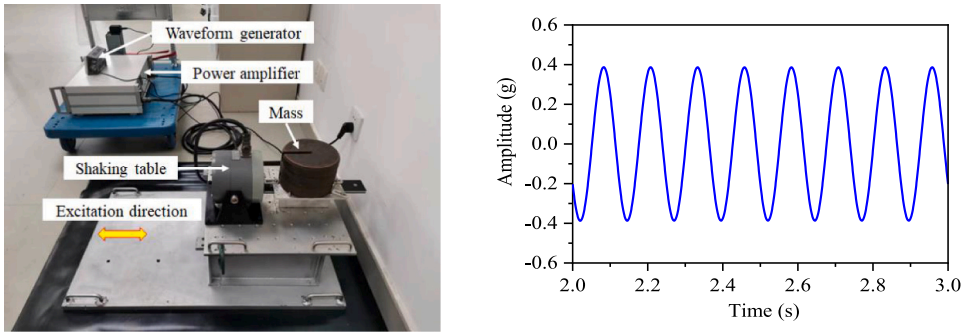


Fig. 15. Time synchronization errors under different synchronization intervals.



(a) Setup of the shaking table (b) Acceleration time history of the mass block

Fig. 16. Shaking table and time history of mass.

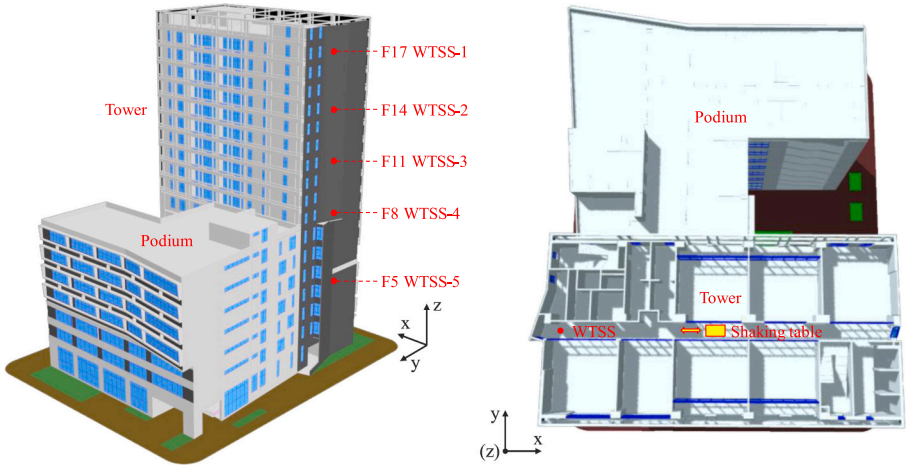


Fig. 17. Arrangement of shaking table and sensors in the building.

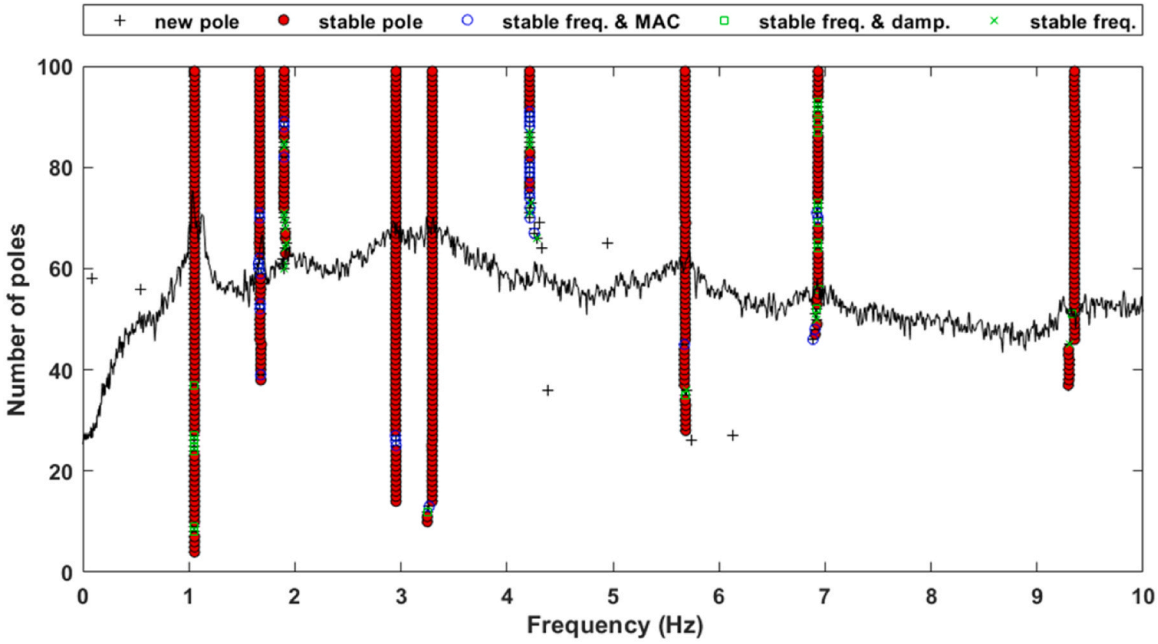


Fig. 18. SSI-COV stabilization diagram for modal identification of the building.

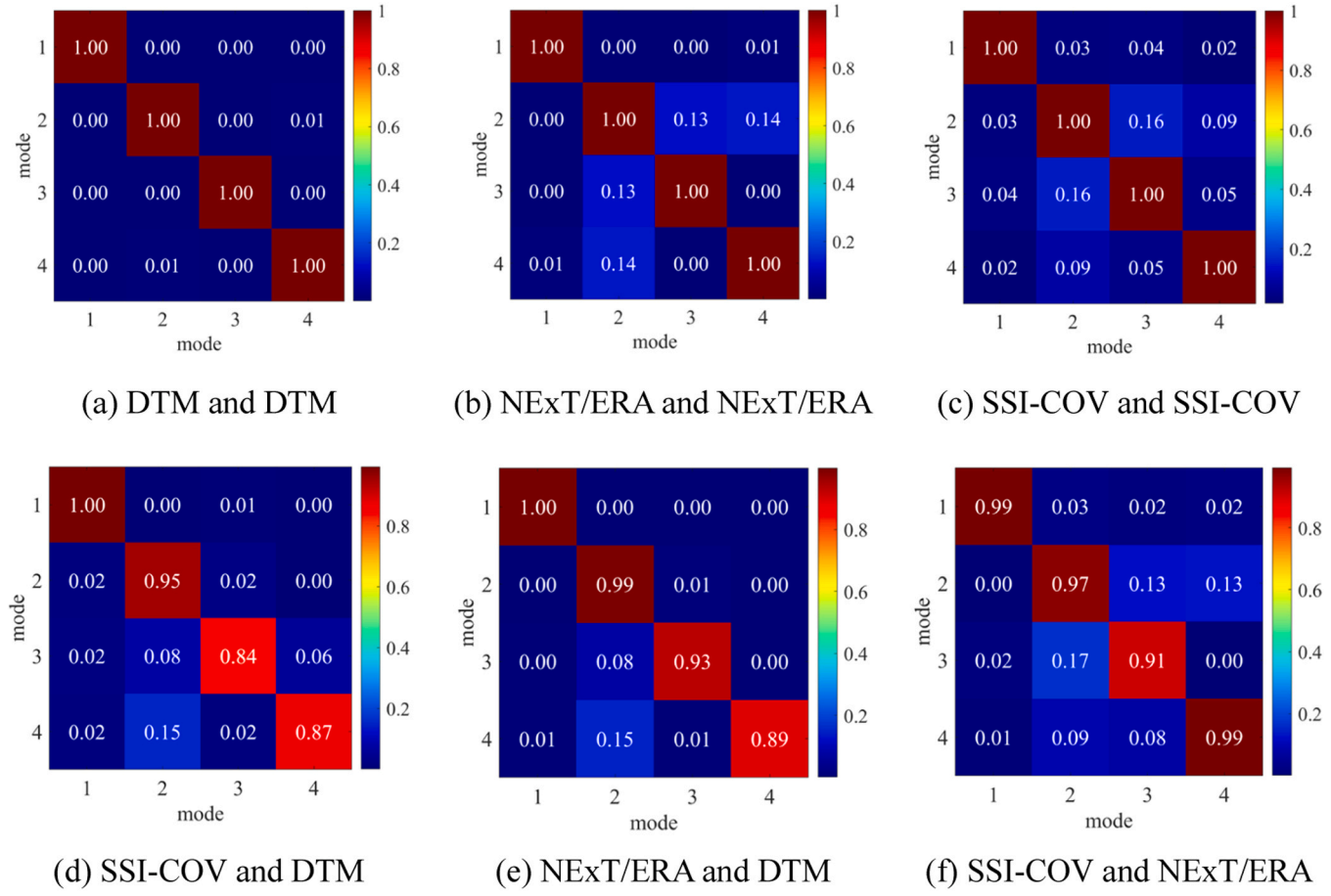


Fig. 19. Modal assurance criterion among different models.

mode shapes (Fig. 19c) [12]. Furthermore, the retained poles were then cross-checked against the calibrated digital-twin model (Fig. 19d). Most diagonal MAC values exceed 0.85 while the off-diagonal terms remain near zero, confirming that the experimentally-identified modes are physically meaningful and mutually orthogonal, rather than numerical artefacts.

For independent verification, the same data set was also processed with the NExT-ERA technique [64,65]. A comprehensive correlation study among the three data sets (DTM, SSI-COV and NExT/ERA) is presented in Fig. 19, which assembles six MAC heat maps. The auto-MAC of the DTM (Fig. 19a) exhibits unit diagonals and negligible off-diagonal entries, verifying the mathematical orthogonality of the numerical eigenvectors. Auto-MAC matrices for NExT/ERA and SSI-COV (Figs. 19b and 19c) maintain unit diagonals and off-diagonals below 0.16, demonstrating that measurement noise and order selection have only a minor influence on shape orthogonality. Cross-MAC analysis further substantiates the robustness of the experimental procedures: SSI-COV versus DTM (Fig. 19d) yields diagonal MACs of 1.00, 0.95, 0.84 and 0.87 for modes 1–4, whereas NExT/ERA versus DTM (Fig. 19e) attains 1.00, 0.99, 0.93 and 0.89. The direct comparison between the two data-driven techniques (Fig. 19f) reveals almost indistinguishable mode shapes, with diagonals of 0.99, 0.97, 0.91 and 0.99 and all off-diagonals below 0.17. Because MAC values exceeding 0.80 are widely accepted as indicative of strong correlation and those above 0.90 as excellent, the uniformly high diagonals and low off-diagonals across all matrices provide compelling quantitative evidence that the poles retained after the SSI-COV stability screening correspond to the structure's true dynamic characteristics and that the three identification routes are internally consistent and mutually reinforcing.

After eliminating the false modes, the first to fourth order natural frequencies along the x-axis are shown in Table 1 and the first to fourth order mode shapes are shown in Fig. 20.

Table 1
Four natural frequencies along the x-axis obtained from test and modal analysis.

	Natural frequency (Hz)		
Mode	Digital twin model	SSI-COV	NExT/ERA
Mode-1	0.979	1.054	1.036
Mode-2	3.222	2.892	2.950
Mode-3	6.095	6.916	6.969
Mode-4	9.838	9.337	9.242

The MDOF flexural-shear model described in Section 4.1 is used as the structural dynamics DTM for the building. The mass of each floor is determined according to the design details, and the stiffness of the flexural and shear springs were updated using the particle swarm optimization algorithm (PSO) according to the modal parameters obtained through operational modal analysis. The first to fourth order modes of the updated DTM agree well with the test results, as shown in Table 1 and Fig. 20.

6.2.2. *Detection of abnormal vibration*

In the case study, the historical spectrum envelopes of all WTSS nodes were acquired through the short-time Fourier transform (STFT) of the collected vibration data of the building under normal operation, as shown by the dashed lines in Fig. 21a. The abnormal vibration is detected if the current vibration spectrum amplitude exceeds twice that of the historical spectrum envelope at some specific frequency point.

The excitation test started with the shaking table applying an 8 Hz simple harmonic excitation on the 5th floor of the building. As depicted

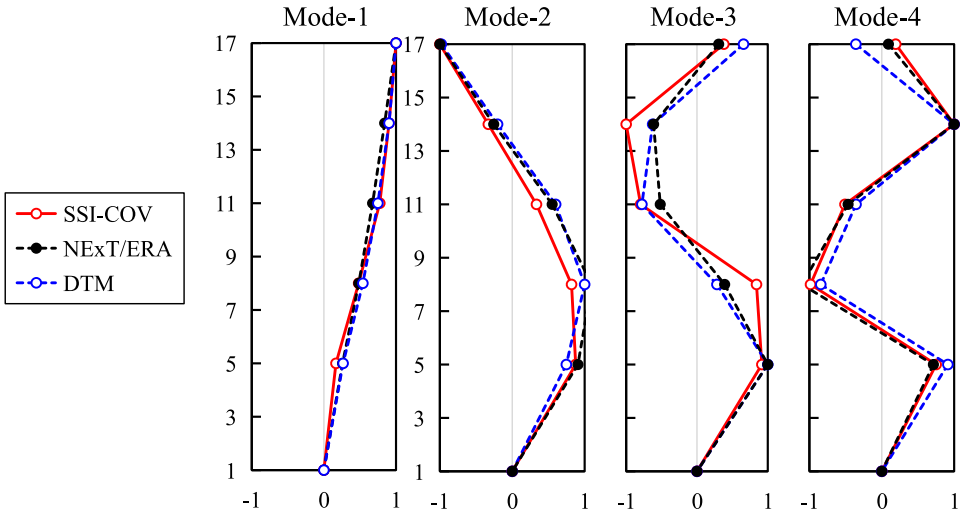
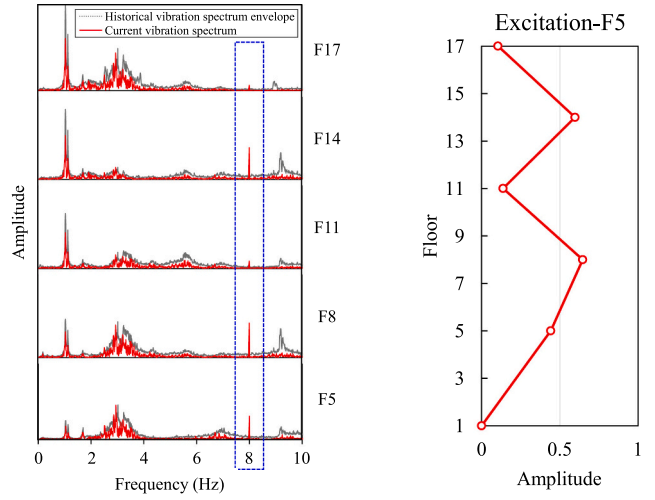


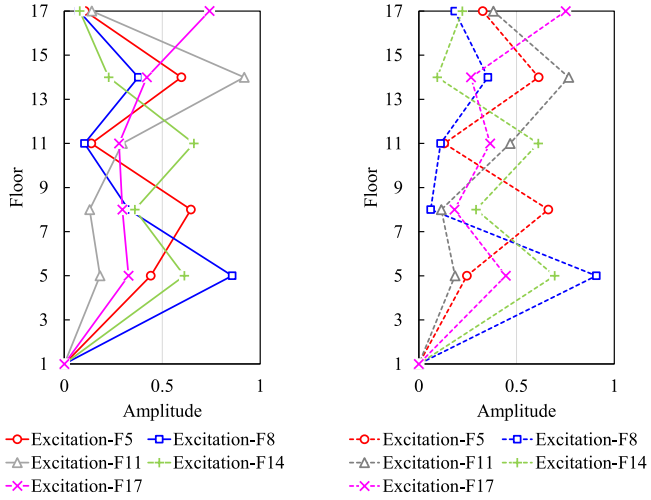
Fig. 20. Mode shapes obtained from OMA test and modal analysis of the digital twin model.



(a) Acceleration spectra and vibration anomaly detection

(b) SAF vector

Fig. 21. Vibration anomaly detection and SAF vector with 8 Hz excitation on the 5th floor.



(a) Shaking table excitations

(b) DTM simulations

Fig. 22. SAF vectors obtained through shaking table excitation test and DTM simulation with excitations on the 5th, 8th, 11th, 14th and 17th floors respectively.

by the red curve in Fig. 21a, all WTSS nodes were able to detect the vibration anomaly at 8 Hz. For instance, even though the node located on the 17th floor is considerably distant from the excitation source on the 5th floor, it can still capture the 8 Hz spike. It is worth noting that the vibration generated by the shaking table was very subtle (with an excitation force of 82.9 N), and completely imperceptible to human. The clearly identified 8 Hz spikes on all five floors indicate the high sensitivity of the WTSS nodes. The normalized vibration anomaly SAF vector is calculated by the cloud server as shown in Fig. 21b. The excitation tests were further carried out on the 8th, 11th, 14th and 17th floors subsequently, and the corresponding SAF vectors are shown in Fig. 22a.

6.2.3. *Localization of vibration source*

According to the vibration anomaly frequency of 8 Hz identified in Section 6.2.2, a series of time-history analyses were performed using the DTM with the 8 Hz sinusoidal excitation applied on the 5th, 8th, 11th, 14th and 17th floors respectively, and the corresponding SAF vectors are shown in Fig. 22b.

The measured vibration anomaly SAF matrix V (i.e. $[v_5, v_8, v_{11}, v_{14}, v_{17}]^T$) and the simulated SAF matrix W for excitation source on the 5th, 8th, 11th, 14th and 17th floors were obtained respectively through the preceding excitation tests and time-history analyses. The similarity matrix S (i.e. $[s_5, s_8, s_{11}, s_{14}, s_{17}]^T$) was calculated using the equation $S = V \times W$ to demonstrate the similarity between the measured and simulated SAF vectors, as shown in Fig. 23. The first row of the similarity matrix S is s_5, which means the similarity between the measured vibration anomaly SAF vector v_5 (excitation on the 5th floor) and the simulated SAF matrix W. The maximum value of s_5 is the fifth value, which means the measured SAF vector s_5 exhibits the best similarity with the simulated SAF vector w_6 (excitation applied on the 6th floor). This indicates that the predicted floor location of the vibration source has a one-floor margin of error. Considering that the primary objective of this study is to achieve zonal-level vibration source localization rather than floor-specific identification, this limitation is deemed acceptable for the current research scope.

To better demonstrate the performance of edge and cloud computing paradigms, we separately recorded the analysis times for both

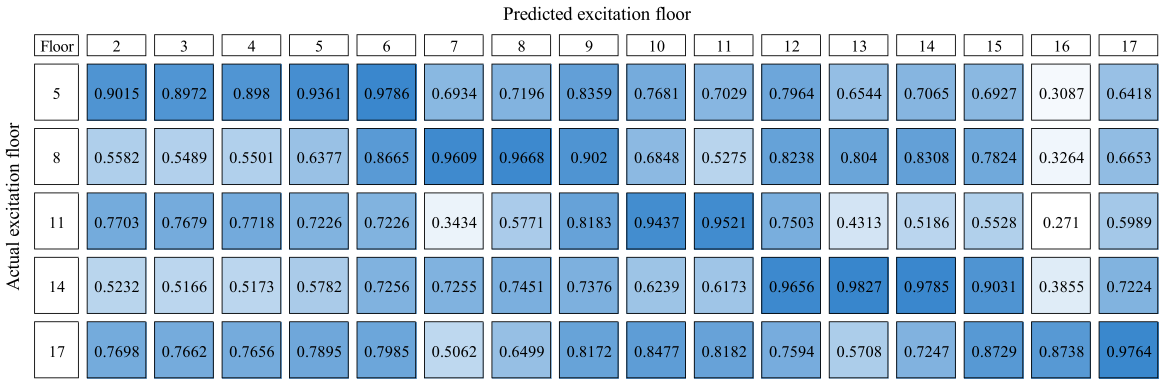


Fig. 23. Similarity matrix S between the measured SAF matrix V and the simulated SAF matrix W.

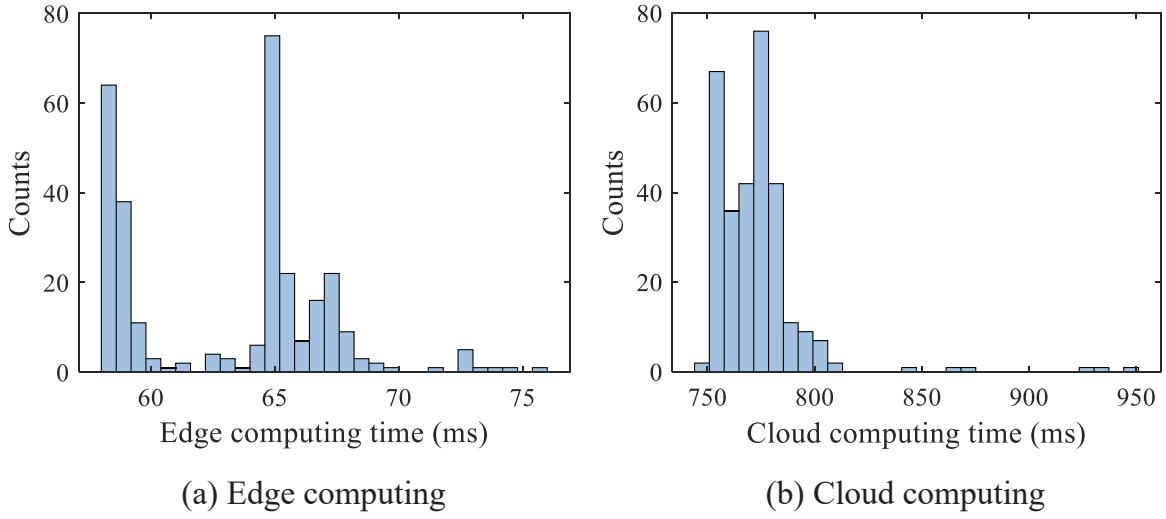


Fig. 24. Computational performance of edge and cloud computing.

approaches. Edge computing was implemented on a Raspberry Pi Zero 2 W module, performing the vibration anomaly detection algorithm. Through 300 test runs, edge computing consistently achieved processing times below 80 ms (Fig. 24a). For cloud computing, the tests were conducted on a server equipped with an Intel i5–9600KF CPU and 48 GB RAM. After 300 test runs, all computational times were less than 1 s (Fig. 24b).

6.2.4. *Visualization platform*

This study developed a web-based platform for the visualization of vibration data and assessment results. Upon completion of the analysis regarding vibration anomaly detection performed by the back-end, the obtained results are subsequently transmitted to the front-end webpage for visualization. As shown in Fig. 25a, the platform demonstrates an alert of a detected vibration anomaly on the 17th floor of the building. Furthermore, the platform provides detailed information concerning the position and the status of the WTSS nodes (Fig. 25b). Moreover, the platform provides real-time visualization of acceleration time-history and frequency spectrum of each sensor node, as shown in Fig. 25c. In summary, this web-based visualization platform greatly facilitates the vibrational safety management and assessment of the building.

7. Discussion and limitations

This study presents an integrated wireless smart sensor system with edge-cloud computing algorithms for vibration monitoring in high-rise buildings, featuring three key innovations. First, we developed an edge-cloud collaborative framework that distributes vibration detection algorithms to edge nodes, which reduces cloud computing workload and data transmission requirements while enabling near real-time vibration source localization and visualization. Second, we designed a compact, highly integrated wireless sensor that combines high-precision vibration measurement (0.2 μg/(Hz)$^{1/2}$ noise density) with onboard edge computing capabilities (Linux-based Python environment), achieves less than 20 μs time synchronization through a novel LoRa-based chain network, and enables direct 4 G cloud connectivity without requiring gateway nodes. Third, we established a digital twin-based vibration source localization methodology employing an updated flexural-shear MDOF model for building simulation, performing dynamic analysis under floor-specific excitations, and implementing SAF vector matching between simulated and measured responses to identify vibration sources.

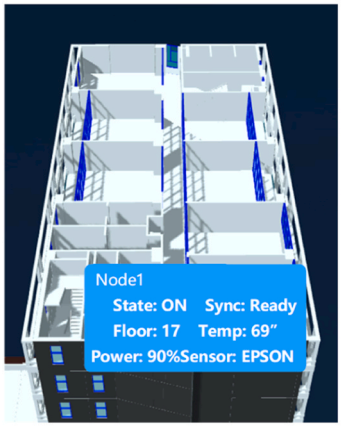
This study demonstrates the feasibility of an edge-cloud collaborative framework and proposes a novel vibration source localization method. However, three key limitations warrant discussion to guide future research directions:

(1) The current SAF vector comparison method achieves only zonal-level localization because SAF vectors exhibit similar patterns when excitations are applied to adjacent floors. This limitation underscores the need for more precise floor-specific identification techniques.

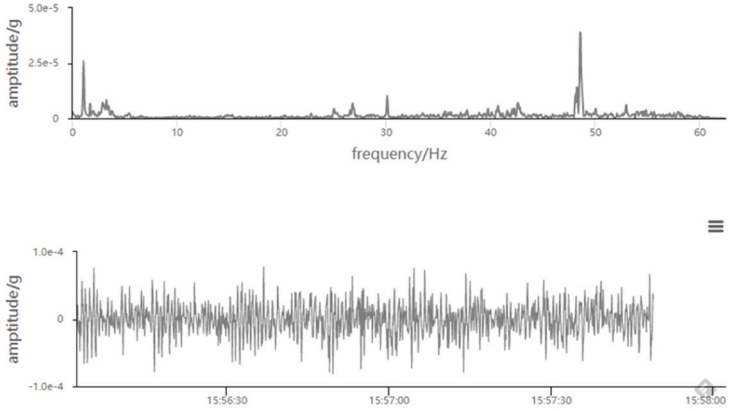
(2) The framework currently supports only single-frequency steady-state excitations. Extending it to multi-source and non-stationary vibration scenarios will require further theoretical and experimental development.



(a) Management platform demonstrating a vibration anomaly alert on the 17th floor.



(b) Visualization of sensor status



(c) Real-time display of acceleration time-history and frequency spectrum

Fig. 25. Illustration of the visualization platform.

(3) Edge sensors currently implement only basic signal processing. Advances in edge computing hardware are expected to enable the deployment of sophisticated deep learning models, thereby improving detection accuracy and optimizing computational resource allocation across the edge-cloud system.

8. Conclusions

An edge-cloud collaborative digital twin framework for vibration anomaly detection and localization in high-rise buildings is proposed in this study. The framework includes a wireless time-synchronized smart sensor (WTSS), a series of vibration anomaly detection and localization methods, and a BIM-empowered online platform. The inferred conclusions are as follows:

(1) A digital twin framework for vibration monitoring of high-rise buildings is proposed in this study. The framework features vibration monitoring and anomaly detection at the edge sensors. Meanwhile, the digital twin model updating, vibration source localization, and platform management are performed at the cloud server. The assessment results can be visualized through a BIM-based management platform.

(2) A wireless time-synchronized smart sensor (WTSS) is proposed in this study. The sensor integrates a high-sensitive accelerometer, an edge computing module, a wireless data transmission module and a wireless time synchronization module. The WTSS sensor is very compact in size with the dimension of 10 cm*8 cm*4.5 cm. The sensitive and vibration measurement performance of the WTSS sensor is validated by comparing with the PCB393B31 piezoelectric sensor.

(3) An edge-cloud collaborative assessment method for vibration anomaly detection and localization in high-rise buildings is proposed. The method was validated through a series of harmonic excitation tests for a 17-story high-rise teaching building. A very weak 8 Hz harmonic excitation with the excitation force amplitude of 82.9 N was applied using a shaking table deployed on 5th, 8th, 11th, 14th, and 17th floors respectively. Although the excitation is very weak, the proposed WTSS sensors can clearly identify the features of the excitation and the vibration anomaly event can be detected through the edge computing of

WTSS sensors using the proposed anomaly detection algorithm. Moreover, the location of the vibration source can be estimated through the cloud computing of the proposed vibration localization algorithm using the structural dynamics digital twin model.

In summary, the outcomes of this study are expected to provide a useful tool for the rapid detection and localization of vibration anomaly in high-rise buildings and assist in safety operation and maintenance of building structures.

CRediT authorship contribution statement

Linlin Xie: Supervision. **Weizong Lai:** Software. **Liu Mei:** Supervision. **Huijin Zhu:** Validation, Writing – original draft, Methodology, Visualization. **Chen Xiong:** Project administration, Funding acquisition, Data curation, Methodology, Formal analysis, Conceptualization. **Kuangyu Chen:** Writing – review & editing, Project administration, Supervision. **Jie Huang:** Writing – original draft, Validation, Visualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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